# Bihar Engineering University, Patna

## B.Tech 4th Semester Examination, 2026

*Course: B.Tech*
*Code: 101406*

*Subject: Engineering Geology*

*Time: 03 Hours*
*Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. 1 is compulsory.*

**Q.1 *Choose the correct option / answer the following (Any seven question only):*** [2 x 7 = 14]

(a) According to Darcy's law, groundwater discharge through saturated rock or soil is directly proportional
(i) Hydraulic gradient and cross-sectional area
(ii) Sample length alone
(iii) Rock colour and mineral composition
(iv) Atmospheric pressure alone

(b) A rock formation has a hydraulic conductivity of $2x10^{-4}$ m/s, a hydraulic gradient of 0.5 and a cross-sectional area of 4 $m^2$. Calculate the groundwater discharge
(i) $2x10^{-4}$ $m^3/s$
(ii) $6x10^{-4}$ $m^3/s$
(iii) $4x10^{-4}$ $m^3/s$
(iv) $8x10^{-4}$ $m^3/s$

(c) The difference in hydraulic head between two points is 6 m, and the distance between them is 30 m. Calculate the hydraulic gradient
(i) 0.10
(ii) 0.30
(iii) 0.20
(iv) 0.50

(d) A drill-core run has a total length of 1.5 m. The combined length of qualifying sound core pieces, each at least 100 mm long, is 1.2 m. Calculate the RQD
(i) 60%
(ii) 80%
(iii) 70%
(iv) 90%

(e) Identification of crystalline mineral phases in a rock sample is commonly performed using
(i) X-ray diffraction
(ii) Slump testing
(iii) Sieve analysis
(iv) Liquid limit testing

(f) Which seismic waves generally arrive first at a seismic recording station?
(i) S-waves
(ii) P-waves
(iii) Surface waves
(iv) Tsunami waves

(g) Rock Quality Designation is primarily used to assess
(i) The colour of a rock mass
(ii) The chemical composition of groundwater
(iii) The quality of a rock mass using drill-core recovery information
(iv) The age of a mineral deposit directly

(h) The angle that a geological plane makes with the horizontal, measured in the direction of maximum inclination, is called
(i) Strike
(ii) Outcrop
(iii) Dip
(iv) Fold axis



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(i) According to Darcy's law, the discharge through a saturated soil or rock sample is directly proportional to
(i) Hydraulic gradient and cross-sectional area
(ii) Total length of the sample only
(iii) Porosity alone
(iv) Atmospheric temperature only

(j) Which of the following techniques is commonly used to study the crystalline phases present in a rock or mineral sample?
(i) X-ray diffraction
(ii) Slump test
(iii) Sieve analysis
(iv) Liquid limit test

**Q.2** (a) Explain the significance of geological investigations in the selection and design of foundations for major civil engineering structures. [7]

(b) Discuss the applications of SEM and XRD in the investigation and identification of minerals and rocks. [7]

**Q.3** (a) Describe the physical properties commonly used for the identification of minerals, with suitable examples. [7]

(b) Differentiate between physical and optical mineralogy, highlighting their importance in engineering geological investigations. [7]

**Q.4** (a) Explain the rock cycle with a neat, labelled diagram, showing the processes responsible for the formation of igneous, sedimentary and metamorphic rocks. [7]

(b) Compare the formation, texture and engineering characteristics of granite, sandstone and gneiss. [7]

**Q.5** (a) Classify igneous rocks based on their mode of occurrence and explain the characteristics of intrusive and extrusive rocks with examples. [7]

(b) Describe the formation of sedimentary rocks and discuss the engineering significance of bedding, porosity and cementation. [7]

**Q.6** (a) During a borehole investigation for a proposed tunnel, the total core-run length is 2.0 m. The recovered sound core pieces measuring at least 100 mm are 0.30 m, 0.25 m, 0.40 m, 0.35 m and 0.20 m. Calculate the RQD and classify the rock quality using the conventional RQD classification. [7]

(b) If additional qualifying core pieces with a combined length of 0.20 m are obtained from a separate 2.0 m core run, calculate the revised RQD for that run and compare the two results. [7]

**Q.7** (a) Water flows through a saturated rock mass having a hydraulic conductivity of $4 \times 10^{-5}$ m/s. The cross-sectional area normal to the flow is 30 $m^2$, and the measured discharge is $3.6 \times 10^{-4}$ $m^3/s$. Calculate the hydraulic gradient through the rock mass. [7]

(b) If the distance between the two observation points is 25 m, determine the difference in hydraulic head. [7]

**Q.8** (a) Classify mass-movement processes and explain the factors responsible for landslides in natural and engineered slopes. [7]

(b) Discuss suitable preventive and mitigation measures for landslides and describe the potential effects of volcanic activity on civil engineering works. [7]

**Q.9** Write short note on ***any two*** of the following: [7 x 2 = 14]
(a) Types of Geological Faults
(b) Hydrological Cycle
(c) Earthquake Magnitude and Intensity
(d) Types and Causes of Landslides



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